ia_study1_mainAnalyses

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```r
#setwd(dirname(rstudioapi::getActiveDocumentContext()$path))
library(tidyverse)
library(emmeans)
library(psych)
library(patchwork)
source("https://raw.githubusercontent.com/vyqlua/avi_functions/refs/heads/main/ipsatize_fn.R")
```

```r
sensitivity_analyses <- c("!is.na(subID)",
  "exclude_age != 1", "exclude_sd != 1",
  "exclude_age != 1 & exclude_sd != 1")
sensitivity_analyses_i <- c("!is.na(subID)", "exclude_age != 1")
```

Load Data

```
data <- read_csv("ia_study1_anon_data_computed.csv")
data <- data[ , -1]

levelBreakdown <- read_csv("ia_TrialsBreakdown.csv")
```

Are There Cultural Differences in Ideal Affect between European Americans and Taiwanese?

```
#plot_avi(data, group_id = "cultureEA")
#need to figure out what affective states to include in AVI measure
```

Does Induced Mood Influence Giving Towards Excited, Calm, and Neutral Responders Differently Across Cultures?

Admin before analyses

```
# Define the analysis function

aov_analysis <-
  function(formula, data, filter_condition1, filter_condition2 = NULL)
    {
    # Initialize a list to store results from each iteration
    analysis_result <- list()
  if (is.null(filter_condition2)) {
    for (i in 1:length(filter_condition1)) {
      subset_data <- data %>%
        filter(eval(parse(text = filter_condition1[i])))
    # Run the analysis on the subset
    analysis_result[[filter_condition1[i]]] <-
      aov(formula, data = subset_data) %>% summary()
    }}
  else {
    for (j in 1:length(filter_condition2)) {
      for (i in 1:length(filter_condition1)) {
        subset_data <- data %>%
          filter(eval(parse(text = filter_condition1[i]))) %>%
          filter(eval(parse(text = filter_condition2[j])))
    # Run the analysis on the subset
    analysis_result[[
      paste(filter_condition1[i], "&", filter_condition2[j])]] <-
      aov(formula = formula, data = subset_data) %>% summary()
      }}
    }
  return(analysis_result)
}
```

```
# convert AVI data into long form
longGivingdata_i <- data %>%
  dplyr::select(
    subID, cultureEA, order,
    starts_with("raw") & contains("_i"),
    starts_with("exclude"),
    age # covariates
    ) %>%
  pivot_longer(
    cols = `raw1_i`:`raw36_i`,
    names_to = "trial",
    values_to = "response_i") %>%
  dplyr::mutate(
    trial = as.numeric(substr(trial, start = 4, stop = str_length(trial)-2)))
```

```r
# add info on incidental affect (mood), face gender, face race, and face emotion for each trial.
longGivingdata_i = longGivingdata_i %>%
  dplyr::mutate(
    mood = levelBreakdown$mood[trial],
    faceGender = levelBreakdown$faceGender[trial],
    faceRace = levelBreakdown$faceRace[trial],
    faceEmotion = levelBreakdown$faceEmotion[trial])

# make variables factors
longGivingdata_i = longGivingdata_i %>%
  dplyr::mutate(
    culture = ifelse(cultureEA == 1, "EA", "TN") %>% as.factor(),
    mood = factor(mood, levels = c("HAP", "HAN","Neutral")),
    faceRace = factor(faceRace, levels = c("White", "Asian")),
    faceGender = factor(faceGender, levels = c("Male", "Female")),
    faceEmotion = factor(faceEmotion, levels = c("Excited", "Calm", "Neutral")),
    subID = factor(subID, unique(subID)))

longGivingdata_i = longGivingdata_i %>% dplyr::filter(!is.na(response_i))
```

Sample size

Main analyses (!is.na(subID))

```r
print("n_total:"); longGivingdata_i %>%
  dplyr::summarise(n = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_total:"
```

```
## [1] 184
```

```r
print("n_tw:"); longGivingdata_i %>%
  dplyr::filter(culture == "TN") %>%
  dplyr::summarise(n_tw = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_tw:"
```

```
## [1] 107
```

```r
print("n_ea:"); longGivingdata_i %>%
  dplyr::filter(culture == "EA") %>%
  dplyr::summarise(n_ea = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_ea:"
```

```
## [1] 77
```

Supple analyses (exclude_age != 1)

```
print("n_total:"); longGivingdata_i %>%
  dplyr::filter(exclude_age != 1) %>%
  dplyr::summarise(n = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_total:"
```

```
## [1] 175
```

```
print("n_tw:"); longGivingdata_i %>%
  dplyr::filter(exclude_age != 1, culture == "TN") %>%
  dplyr::summarise(n_tw = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_tw:"
```

```
## [1] 106
```

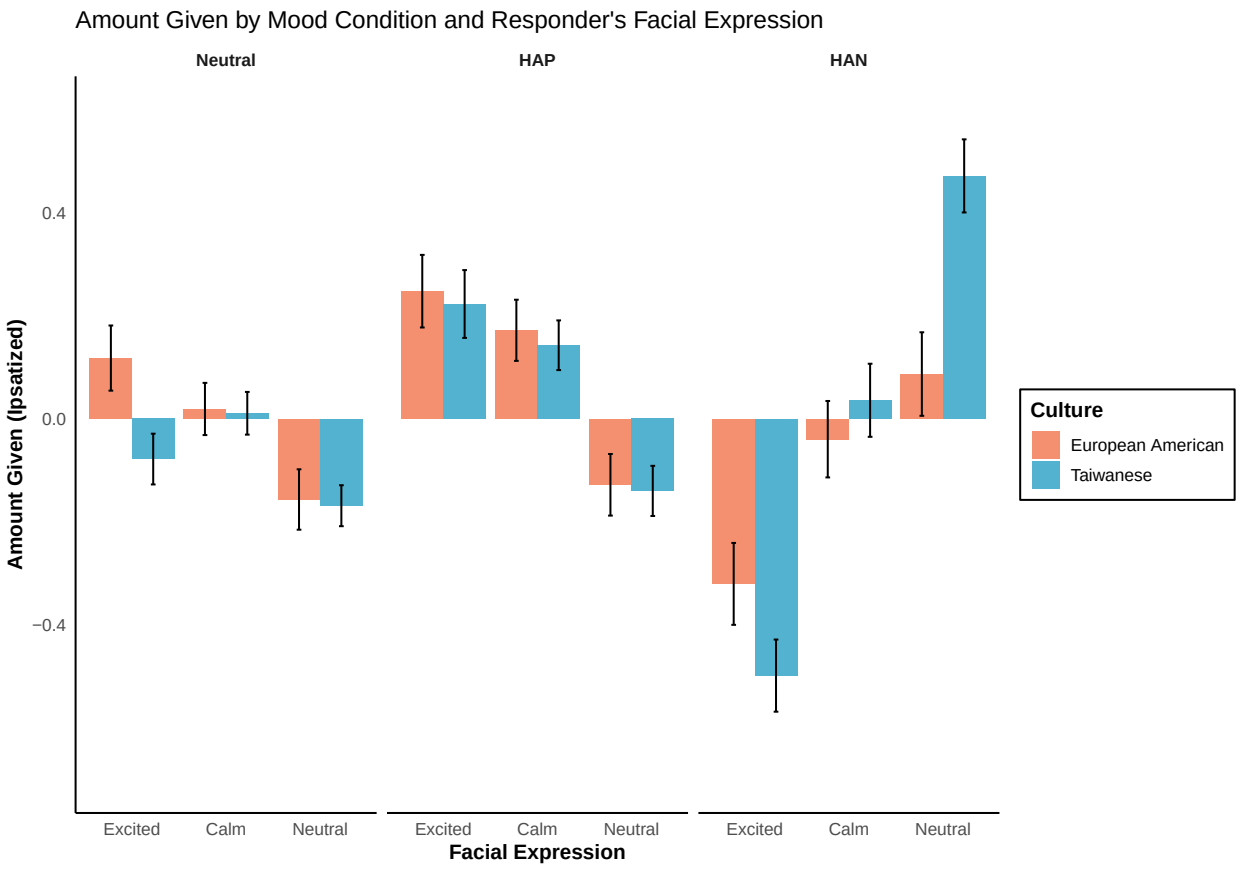
```
print("n_ea:"); longGivingdata_i %>%
  dplyr::filter(exclude_age != 1, culture == "EA") %>%
  dplyr::summarise(n_ea = dplyr::n_distinct(subID)) %>%
  pull()
```

```
## [1] "n_ea:"
```

```
## [1] 69
```

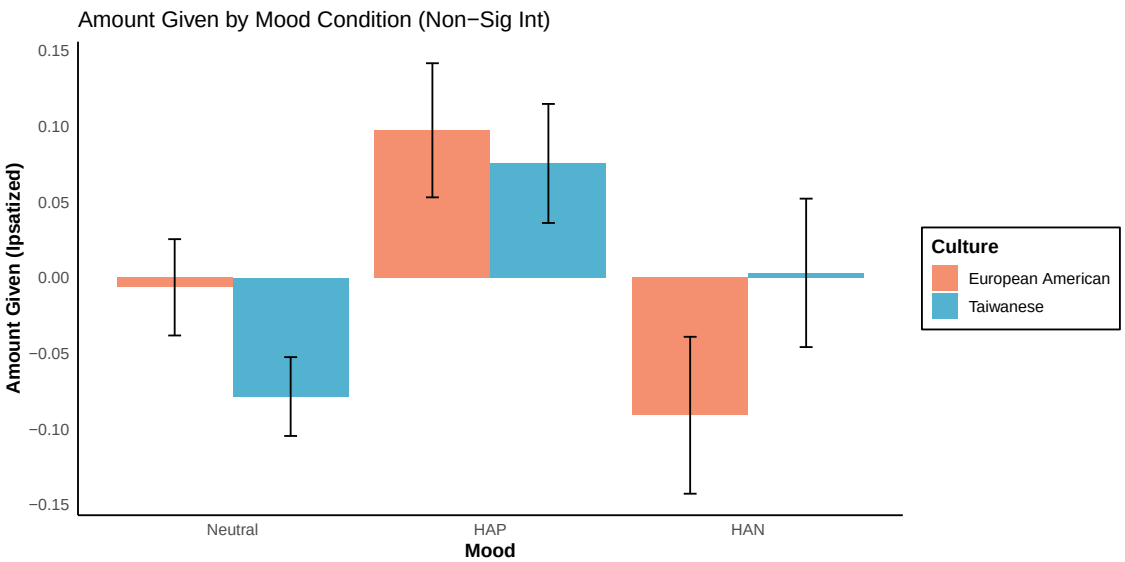
Visualizations

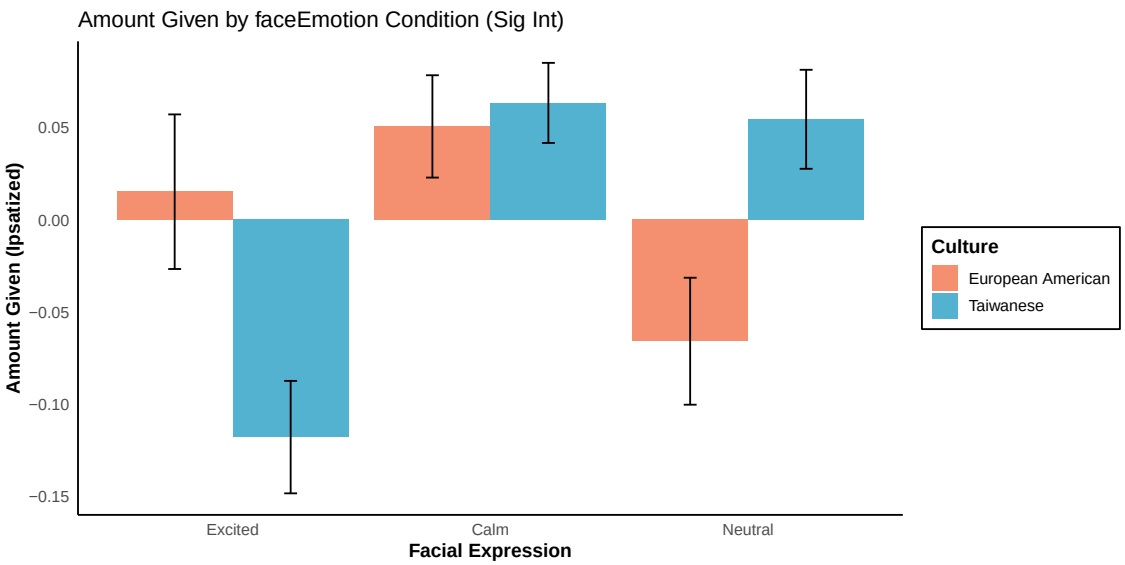
```
longGivingdata_i2 <- longGivingdata_i %>% dplyr::mutate(
  culture = if_else(culture == "EA", "European American", "Taiwanese")) %>%
  dplyr::rename(Culture = culture) %>%
  dplyr::mutate(
    mood = factor(mood,
                  labels = c("Neutral", "HAP", "HAN"),
                  levels = c("Neutral", "HAP", "HAN")))
ggplot(
  longGivingdata_i2 %>%
    group_by(subID, mood, faceEmotion, Culture) %>%
    summarise(response_i = mean(response_i, na.rm = T)) %>%
    group_by(faceEmotion, mood, Culture) %>%
    summarise(mean = mean(response_i, na.rm = T),
              SD = sd(response_i, na.rm = T),
              SE = sd(response_i, na.rm = T)/sqrt(n())),
  aes(faceEmotion, mean, fill = Culture)) +
  facet_grid(~mood)+
  scale_fill_manual(
    values = c("European American"="#F49070", "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1)+
  xlab("Facial Expression") +
  ylab("Amount Given (Ipsatized)") +
  ylim(c(-0.7, 0.6)) +
  ggtitle("Amount Given by Mood Condition and Responder's Facial Expression")
```

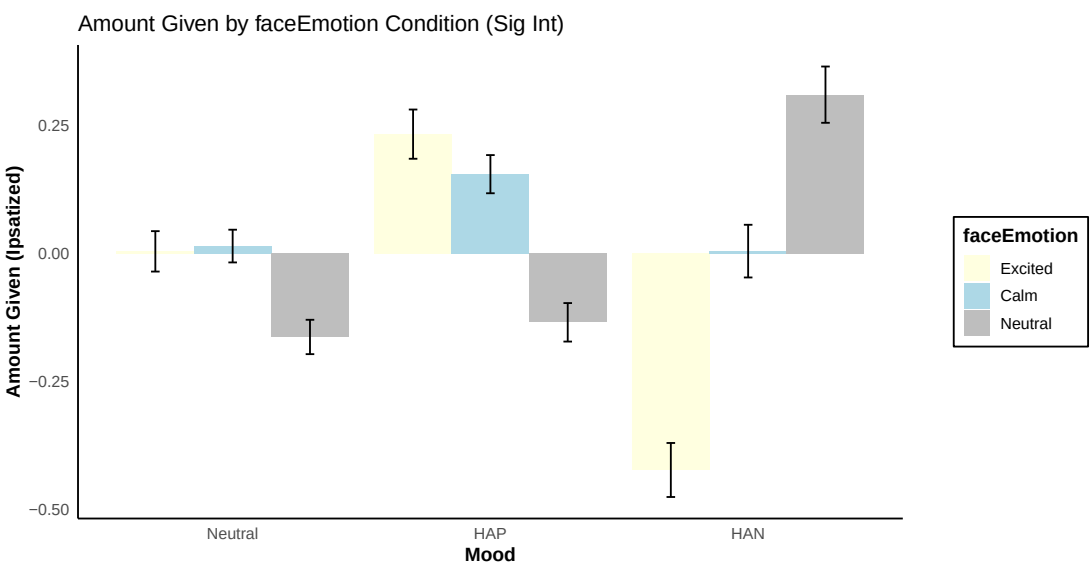

Amount Given by Mood Condition and Responder's Facial Expression
Neutral
HAP
HAN
0.4
0.0
−0.4
Amount Given (Ipsatized)
Excited
Calm
Neutral
Excited
Calm
Neutral
Excited
Calm
Neutral
Facial Expression
Culture
European American
Taiwanese

```r
p1 <- ggplot(longGivingdata_i2 %>%
               group_by(subID, mood, Culture) %>%
               summarise(response_i = mean(response_i, na.rm = T)) %>%
               group_by(mood, Culture) %>%
               summarise(
                 mean = mean(response_i, na.rm=T),
                 SD = sd(response_i, na.rm = T),
                 SE = sd(response_i, na.rm = T)/sqrt(n())),
             aes(mood, mean, fill = Culture)) +
  scale_fill_manual(values =
                      c("European American"="#F49070",
                        "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Mood") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by Mood Condition (Non-Sig Int)")

p2 <- ggplot(longGivingdata_i2 %>%
               group_by(subID, faceEmotion, Culture) %>%
               summarise(response_i = mean(response_i, na.rm = T)) %>%
               group_by(faceEmotion, Culture) %>%
               summarise(
                 mean = mean(response_i, na.rm=T),
                 SD = sd(response_i, na.rm = T),
                 SE = sd(response_i, na.rm = T)/sqrt(n())),
             aes(faceEmotion, mean, fill = Culture)) +
  scale_fill_manual(values =
                      c("European American"="#F49070",
                        "Taiwanese"="#52B2CF")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
                position = position_dodge(0.9), width = 0.1) +
  xlab("Facial Expression") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by faceEmotion Condition (Sig Int)")

p3 <- ggplot(longGivingdata_i2 %>%
               group_by(subID, faceEmotion, mood) %>%
               summarise(response_i = mean(response_i, na.rm = T)) %>%
               group_by(faceEmotion, mood) %>%
               summarise(
                 mean = mean(response_i, na.rm=T),
                 SD = sd(response_i, na.rm = T),
                 SE = sd(response_i, na.rm = T)/sqrt(n())),
             aes(mood, mean, fill = faceEmotion)) +
  scale_fill_manual(values =
                      c("Neutral"="grey",
                        "Excited"="lightyellow",
                        "Calm"="lightblue")) +
  geom_bar(stat='identity', position='dodge') +
  geom_errorbar(aes(ymin = mean-SE, ymax = mean+SE),
```

```r
                 position = position_dodge(0.9), width = 0.1) +
  xlab("Mood") +
  ylab("Amount Given (Ipsatized)") +
  ggtitle("Amount Given by faceEmotion Condition (Sig Int)")

p1 + p2 + p3 +
  patchwork::plot_layout(
    ncol = 1
  )
```


Amount Given by Mood Condition (Non−Sig Int)
Amount Given (Ipsatized)
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Neutral
HAP
HAN
Mood
Culture
European American
Taiwanese


Amount Given by faceEmotion Condition (Sig Int)
Amount Given (Ipsatized)
0.05
0.00
−0.05
−0.10
−0.15
Excited
Calm
Neutral
Facial Expression
Culture
European American
Taiwanese


Amount Given by faceEmotion Condition (Sig Int)
Amount Given (Ipsatized)
0.25
0.00
−0.25
−0.50
Neutral
HAP
HAN
Mood
faceEmotion
Excited
Calm
Neutral

Analyses

```r
aov_analysis(
  formula = response_i ~ mood*culture*faceEmotion +
    Error(subID/(mood*faceEmotion)),
  data = longGivingdata_i,
  filter_condition1 = sensitivity_analyses_i)
```

```
## $`!is.na(subID)`
##
## Error: subID
##            Df    Sum Sq   Mean Sq F value Pr(>F)
## culture     1 7.000e-29 6.977e-29   0.318  0.573
## Residuals 182 3.987e-26 2.191e-28
##
## Error: subID:mood
##               Df Sum Sq Mean Sq F value Pr(>F)
## mood           2   23.9  11.926   4.227 0.0153 *
## mood:culture   2    7.8   3.903   1.384 0.2520
## Residuals    364 1027.0   2.821
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:faceEmotion
##                      Df Sum Sq Mean Sq F value  Pr(>F)
## faceEmotion           2   16.0   7.993   5.303 0.00537 **
## culture:faceEmotion   2   17.4   8.680   5.759 0.00345 **
## Residuals           364  548.6   1.507
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:mood:faceEmotion
##                           Df Sum Sq Mean Sq F value  Pr(>F)
## mood:faceEmotion           4  254.5   63.63  64.318 < 2e-16 ***
## mood:culture:faceEmotion   4   15.2    3.81   3.852 0.00418 **
## Residuals                728  720.2    0.99
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: Within
##             Df Sum Sq Mean Sq F value Pr(>F)
## Residuals 4968   3810  0.7668
##
## $`exclude_age != 1`
##
## Error: subID
##            Df    Sum Sq   Mean Sq F value Pr(>F)
## culture     1 1.800e-28 1.784e-28   0.339  0.561
## Residuals 173 9.109e-26 5.266e-28
##
## Error: subID:mood
##               Df Sum Sq Mean Sq F value Pr(>F)
## mood           2   23.7  11.867   4.228 0.0153 *
```

```
## mood:culture   2    4.5   2.240   0.798 0.4511
## Residuals    346  971.2   2.807
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:faceEmotion
##                      Df Sum Sq Mean Sq F value  Pr(>F)
## faceEmotion           2   19.1   9.561   6.161 0.00235 **
## culture:faceEmotion   2   14.1   7.073   4.558 0.01112 *
## Residuals           346  537.0   1.552
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: subID:mood:faceEmotion
##                           Df Sum Sq Mean Sq F value  Pr(>F)
## mood:faceEmotion           4  242.5   60.61  61.089 < 2e-16 ***
## mood:culture:faceEmotion   4   15.9    3.98   4.014 0.00316 **
## Residuals                692  686.6    0.99
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Error: Within
##             Df Sum Sq Mean Sq F value Pr(>F)
## Residuals 4725   3610  0.7641
```

Pairwise

```
pairwise <- aov(
  formula = response_i ~ mood*culture*faceEmotion +
    Error(subID/(mood*faceEmotion)),
  data = longGivingdata_i)
```

Main effect mood

```r
# main effect mood
emm_3way_mood <- emmeans(pairwise, ~ mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_mood) # p values
```

```
##  contrast      estimate     SE  df t.ratio p.value
##  HAP - HAN       0.1302 0.0512 364   2.542  0.0307
##  HAP - Neutral   0.1288 0.0512 364   2.515  0.0330
##  HAN - Neutral  -0.0014 0.0512 364  -0.027  0.9996
##
## Results are averaged over the levels of: culture, faceEmotion
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
#confint(pairs(emm_3way_mood)) # confidence intervals
```

Main effect faceEmotion

```r
# main effect faceEmotion
emm_3way_faceEmotion <- emmeans(pairwise, ~ faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_faceEmotion) # p values
```

```
##  contrast          estimate     SE  df t.ratio p.value
##  Excited - Calm     -0.1082 0.0375 364  -2.890  0.0113
##  Excited - Neutral  -0.0456 0.0375 364  -1.218  0.4434
##  Calm - Neutral      0.0626 0.0375 364   1.672  0.2172
##
## Results are averaged over the levels of: mood, culture
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
#confint(pairs(emm_3way_faceEmotion)) # confidence intervals
```

Culture:faceEmotion

```r
emm_3way_faceEmotionCulture <- emmeans(pairwise, ~ culture | faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_faceEmotionCulture) # p values
```

```
## faceEmotion = Excited:
##  contrast estimate     SE  df t.ratio p.value
##  EA - TN    0.1330 0.0432 364   3.075  0.0023
##
## faceEmotion = Calm:
##  contrast estimate     SE  df t.ratio p.value
##  EA - TN   -0.0127 0.0432 364  -0.295  0.7685
##
## faceEmotion = Neutral:
##  contrast estimate     SE  df t.ratio p.value
##  EA - TN   -0.1202 0.0432 364  -2.781  0.0057
##
## Results are averaged over the levels of: mood
```

```r
#confint(pairs(emm_3way_faceEmotionCulture)) # confidence intervals
```

Mood:faceEmotion

```r
emm_3way_faceEmotionMood <- emmeans(pairwise, ~ faceEmotion | mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```r
pairs(emm_3way_faceEmotionMood) # p values
```

```
## mood = HAP:
##  contrast          estimate    SE   df t.ratio p.value
##  Excited - Calm     0.07800 0.057 1046   1.370  0.3574
##  Excited - Neutral  0.36934 0.057 1046   6.485  <.0001
##  Calm - Neutral     0.29135 0.057 1046   5.116  <.0001
##
## mood = HAN:
##  contrast          estimate    SE   df t.ratio p.value
##  Excited - Calm    -0.40767 0.057 1046  -7.158  <.0001
##  Excited - Neutral -0.68870 0.057 1046 -12.093  <.0001
##  Calm - Neutral    -0.28104 0.057 1046  -4.935  <.0001
##
## mood = Neutral:
##  contrast          estimate    SE   df t.ratio p.value
```

```
##  Excited - Calm     0.00495 0.057 1046   0.087  0.9958
##  Excited - Neutral  0.18254 0.057 1046   3.205  0.0040
##  Calm - Neutral     0.17759 0.057 1046   3.118  0.0053
##
## Results are averaged over the levels of: culture
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
#confint(pairs(emm_3way_faceEmotionMood)) # confidence intervals
```

Mood:culture:faceEmotion

```r
emm_3way_bymood <- emmeans(pairwise, ~ culture * faceEmotion | mood)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```r
pairs(emm_3way_bymood) # p values
```

```
## mood = HAP:
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited  0.02486 0.0885 1188   0.281  0.9998
##  EA Excited - EA Calm     0.07592 0.0869 1046   0.874  0.9527
##  EA Excited - TN Calm     0.10493 0.0885 1188   1.186  0.8436
##  EA Excited - EA Neutral  0.37585 0.0869 1046   4.327  0.0002
##  EA Excited - TN Neutral  0.38770 0.0885 1188   4.383  0.0002
##  TN Excited - EA Calm     0.05106 0.0885 1188   0.577  0.9925
##  TN Excited - TN Calm     0.08007 0.0737 1046   1.087  0.8868
##  TN Excited - EA Neutral  0.35099 0.0885 1188   3.968  0.0011
##  TN Excited - TN Neutral  0.36284 0.0737 1046   4.924  <.0001
##  EA Calm - TN Calm        0.02901 0.0885 1188   0.328  0.9995
##  EA Calm - EA Neutral     0.29993 0.0869 1046   3.453  0.0076
##  EA Calm - TN Neutral     0.31178 0.0885 1188   3.525  0.0059
##  TN Calm - EA Neutral     0.27092 0.0885 1188   3.063  0.0272
##  TN Calm - TN Neutral     0.28277 0.0737 1046   3.838  0.0018
##  EA Neutral - TN Neutral  0.01185 0.0885 1188   0.134  1.0000
##
## mood = HAN:
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited  0.17807 0.0885 1188   2.013  0.3354
##  EA Excited - EA Calm    -0.28085 0.0869 1046  -3.233  0.0159
##  EA Excited - TN Calm    -0.35642 0.0885 1188  -4.029  0.0008
##  EA Excited - EA Neutral -0.40735 0.0869 1046  -4.690  <.0001
##  EA Excited - TN Neutral -0.79199 0.0885 1188  -8.953  <.0001
##  TN Excited - EA Calm    -0.45892 0.0885 1188  -5.188  <.0001
##  TN Excited - TN Calm    -0.53449 0.0737 1046  -7.254  <.0001
##  TN Excited - EA Neutral -0.58542 0.0885 1188  -6.618  <.0001
##  TN Excited - TN Neutral -0.97006 0.0737 1046 -13.165  <.0001
##  EA Calm - TN Calm       -0.07557 0.0885 1188  -0.854  0.9570
##  EA Calm - EA Neutral    -0.12650 0.0869 1046  -1.456  0.6922
##  EA Calm - TN Neutral    -0.51114 0.0885 1188  -5.778  <.0001
##  TN Calm - EA Neutral    -0.05093 0.0885 1188  -0.576  0.9926
##  TN Calm - TN Neutral    -0.43557 0.0737 1046  -5.911  <.0001
##  EA Neutral - TN Neutral -0.38464 0.0885 1188  -4.348  0.0002
##
## mood = Neutral:
##  contrast                estimate     SE   df t.ratio p.value
##  EA Excited - TN Excited  0.19603 0.0885 1188   2.216  0.2310
##  EA Excited - EA Calm     0.09879 0.0869 1046   1.137  0.8658
##  EA Excited - TN Calm     0.10714 0.0885 1188   1.211  0.8315
##  EA Excited - EA Neutral  0.27454 0.0869 1046   3.161  0.0201
##  EA Excited - TN Neutral  0.28658 0.0885 1188   3.240  0.0155
##  TN Excited - EA Calm    -0.09724 0.0885 1188  -1.099  0.8818
##  TN Excited - TN Calm    -0.08889 0.0737 1046  -1.206  0.8339
```

```
##  TN Excited - EA Neutral  0.07851 0.0885 1188   0.888  0.9495
##  TN Excited - TN Neutral  0.09055 0.0737 1046   1.229  0.8227
##  EA Calm - TN Calm        0.00835 0.0885 1188   0.094  1.0000
##  EA Calm - EA Neutral     0.17575 0.0869 1046   2.023  0.3296
##  EA Calm - TN Neutral     0.18779 0.0885 1188   2.123  0.2761
##  TN Calm - EA Neutral     0.16740 0.0885 1188   1.892  0.4073
##  TN Calm - TN Neutral     0.17944 0.0737 1046   2.435  0.1451
##  EA Neutral - TN Neutral  0.01204 0.0885 1188   0.136  1.0000
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```r
#confint(pairs(emm_3way_bymood)) # confidence intervals
```

```r
emm_3way_byfaceEmotion <- emmeans(pairwise, ~ culture * mood | faceEmotion)
```

```
## Note: re-fitting model with sum-to-zero contrasts
```

```r
pairs(emm_3way_byfaceEmotion) # p values
```

```
## faceEmotion = Excited:
##  contrast                estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP          0.02486 0.0885 1188   0.281  0.9998
##  EA HAP - EA HAN          0.56821 0.1019  846   5.575  <.0001
##  EA HAP - TN HAN          0.74628 0.0885 1188   8.436  <.0001
##  EA HAP - EA Neutral      0.12987 0.1019  846   1.274  0.7991
##  EA HAP - TN Neutral      0.32590 0.0885 1188   3.684  0.0033
##  TN HAP - EA HAN          0.54335 0.0885 1188   6.142  <.0001
##  TN HAP - TN HAN          0.72142 0.0865  846   8.343  <.0001
##  TN HAP - EA Neutral      0.10501 0.0885 1188   1.187  0.8431
##  TN HAP - TN Neutral      0.30103 0.0865  846   3.482  0.0069
##  EA HAN - TN HAN          0.17807 0.0885 1188   2.013  0.3354
##  EA HAN - EA Neutral     -0.43834 0.1019  846  -4.300  0.0003
##  EA HAN - TN Neutral     -0.24231 0.0885 1188  -2.739  0.0685
##  TN HAN - EA Neutral     -0.61641 0.0885 1188  -6.968  <.0001
##  TN HAN - TN Neutral     -0.42038 0.0865  846  -4.862  <.0001
##  EA Neutral - TN Neutral  0.19603 0.0885 1188   2.216  0.2310
##
## faceEmotion = Calm:
##  contrast                estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP          0.02901 0.0885 1188   0.328  0.9995
##  EA HAP - EA HAN          0.21144 0.1019  846   2.074  0.3017
##  EA HAP - TN HAN          0.13587 0.0885 1188   1.536  0.6410
##  EA HAP - EA Neutral      0.15274 0.1019  846   1.498  0.6654
##  EA HAP - TN Neutral      0.16109 0.0885 1188   1.821  0.4525
##  TN HAP - EA HAN          0.18243 0.0885 1188   2.062  0.3080
##  TN HAP - TN HAN          0.10686 0.0865  846   1.236  0.8191
##  TN HAP - EA Neutral      0.12373 0.0885 1188   1.399  0.7279
##  TN HAP - TN Neutral      0.13208 0.0865  846   1.528  0.6465
##  EA HAN - TN HAN         -0.07557 0.0885 1188  -0.854  0.9570
##  EA HAN - EA Neutral     -0.05870 0.1019  846  -0.576  0.9926
##  EA HAN - TN Neutral     -0.05035 0.0885 1188  -0.569  0.9930
##  TN HAN - EA Neutral      0.01687 0.0885 1188   0.191  1.0000
```

```
##  TN HAN - TN Neutral      0.02522 0.0865  846   0.292  0.9997
##  EA Neutral - TN Neutral  0.00835 0.0885 1188   0.094  1.0000
##
## faceEmotion = Neutral:
##  contrast                estimate     SE   df t.ratio p.value
##  EA HAP - TN HAP          0.01185 0.0885 1188   0.134  1.0000
##  EA HAP - EA HAN         -0.21499 0.1019  846  -2.109  0.2833
##  EA HAP - TN HAN         -0.59963 0.0885 1188  -6.779  <.0001
##  EA HAP - EA Neutral      0.02856 0.1019  846   0.280  0.9998
##  EA HAP - TN Neutral      0.04060 0.0885 1188   0.459  0.9975
##  TN HAP - EA HAN         -0.22684 0.0885 1188  -2.564  0.1070
##  TN HAP - TN HAN         -0.61148 0.0865  846  -7.072  <.0001
##  TN HAP - EA Neutral      0.01671 0.0885 1188   0.189  1.0000
##  TN HAP - TN Neutral      0.02875 0.0865  846   0.332  0.9995
##  EA HAN - TN HAN         -0.38464 0.0885 1188  -4.348  0.0002
##  EA HAN - EA Neutral      0.24355 0.1019  846   2.389  0.1610
##  EA HAN - TN Neutral      0.25559 0.0885 1188   2.889  0.0453
##  TN HAN - EA Neutral      0.62819 0.0885 1188   7.101  <.0001
##  TN HAN - TN Neutral      0.64023 0.0865  846   7.404  <.0001
##  EA Neutral - TN Neutral  0.01204 0.0885 1188   0.136  1.0000
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```
#confint(pairs(emm_3way_byfaceEmotion)) # confidence intervals
```